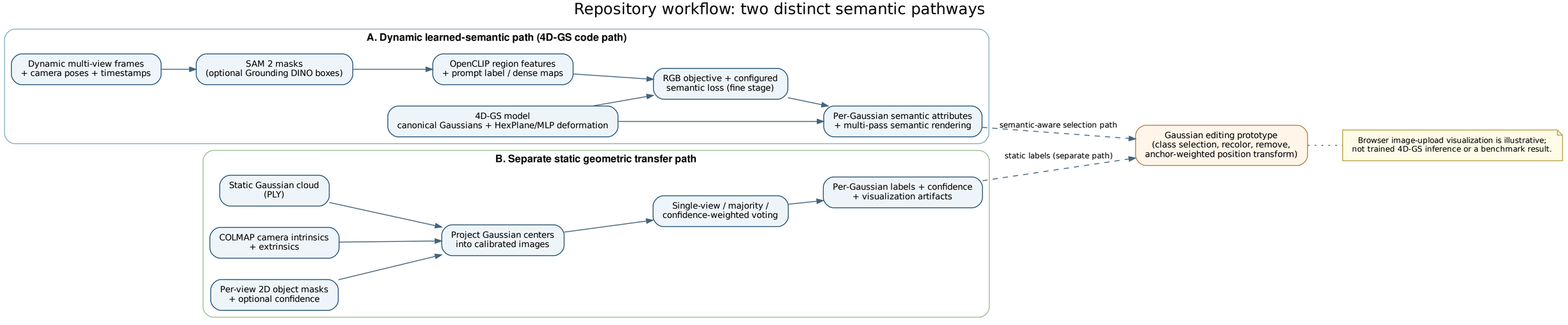
Repository workflow: two distinct semantic pathways
A. Dynamic learned-semantic path (4D-GS code path)
Dynamic multi-view frames
+ camera poses + timestamps
SAM 2 masks
(optional Grounding DINO boxes)
OpenCLIP region features
+ prompt label / dense maps
RGB objective + configured
semantic loss (fine stage)
4D-GS model
canonical Gaussians + HexPlane/MLP deformation
Per-Gaussian semantic attributes
+ multi-pass semantic rendering
B. Separate static geometric transfer path
Static Gaussian cloud
(PLY)
COLMAP camera intrinsics
+ extrinsics
Per-view 2D object masks
+ optional confidence
Project Gaussian centers
into calibrated images
Single-view / majority /
confidence-weighted voting
Per-Gaussian labels + confidence
+ visualization artifacts
semantic-aware selection path
static labels (separate path)
Gaussian editing prototype
(class selection, recolor, remove,
anchor-weighted position transform)
Browser image-upload visualization is illustrative;
not trained 4D-GS inference or a benchmark result.